

# ACTIVATION SPARSIFICATION IN TRANSFORMERS: THRESHOLD PLACEMENT, PRESSURE PLACEMENT, AND SPARSE EXECUTION

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## ABSTRACT

Activation sparsity promises cheaper inference, but interventions that create zeros need not produce execution gains. We isolate pressure, thresholding and their placement through matched Pythia-14M/70M pretraining on MiniPile, connecting validation loss to model-wide activation sparsity, its architectural ceiling, and measured latency. Broad interventions approach the sparsity ceiling at a quality cost. **Local pressure can outperform broad pressure:** at moderate thresholds, FFN-hidden pressure lowers loss in all 16 matched comparisons; across the full threshold grid, it also lowers latency in 18 of 20. It also increases sparsity at unpressured attention outputs. **Execution gains depend on where zeros occur:** in a 14M kernel ablation, skipping FFN-down and attention-output projections reduces latency, whereas joint attention-product skipping increases it. These one-seed results and batch-one RTX5090 measurements support co-designing training interventions and kernels to produce exploitable zero structure.

## 1 INTRODUCTION

Activation sparsity can improve language-model inference efficiency when specialized kernels skip computations associated with zero activations. Recent work has shown that sparsity can be exposed in pretrained models through post-hoc thresholding or induced during training, in both cases with measured inference speedups (Liu et al., 2024; Cetin et al., 2026). These results raise a practical design question: how to effectively sparsify a dense model in activations into an exploitable structure for optimized inference?

Existing approaches induce activation sparsity in different ways. TEAL applies post-hoc clipping to pretrained models, ProSparse uses  $\ell_1$  regularization on feed-forward (FFN) activations during training, and Q-Sparse applies top-k sparsification to both FFN and attention activations (Liu et al., 2024; Song et al., 2024; Wang et al., 2024). While each approach is effective in its own setting, it remains unclear how they compare under a common setup.

To address this gap, we pretrain Pythia-family models from scratch for controlled comparisons across model sizes (Biderman et al., 2023) to study the effects of (i) activation  $\ell_1$  pressure ( $P$ ) and (ii) *thresholding* ( $T$ ). While pressure adds a gradient component to induce smaller activation magnitudes, thresholding maps small values into exact zeros. Each one of them can be placed in different activation sites, giving a combinatorial design set of sparse interventions. For each combination of  $P$  and  $T$ , we derive an architecture-dependent maximum achievable sparsity ( $S_{\text{model}}^{\text{max}}$ ), and measure model-wide attained activation sparsity ( $S_{\text{model}}$ ) and the final inference latency under a specialized sparse kernel. Figure 1 presents sparsity vs. quality trade-offs and inference latency for 14M and 70M Pythia models, across  $P/T$  combinations and post-hoc clipping of baselines.

**High sparsity has a quality cost.** At 14M,  $T_7/P_{\text{all}}$  with  $\kappa = 0.5$  reaches 92% of its ceiling but raises loss by 0.62 over the base model (Figure 3). **Local pressure improves the trade-off.** Compared with  $P_{\text{all}}$ ,  $P_h$  lowers loss by 0.19–0.32 in all 16 matched pairs at  $\kappa \leq 0.1$ , and latency in 18 of 20 pairs overall (Figure 5). **Local pressure has nonlocal effects.** At 14M and  $\kappa = 0.05$ ,  $P_h$  increases unpressured  $z$  sparsity over  $P_0$  under both  $T_4$  and  $T_7$  (Figure 6).

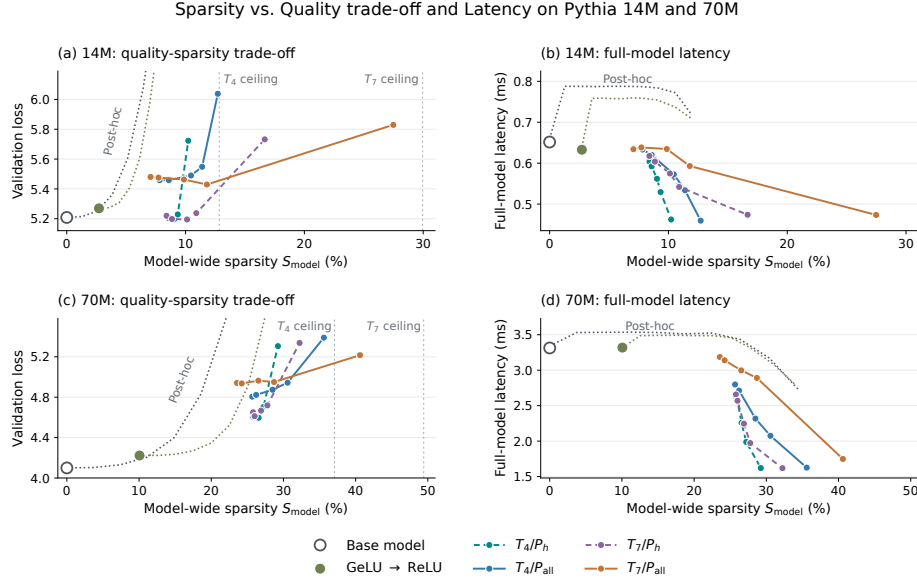


Figure 1: **Quality and execution depend on threshold and pressure placement.** Rows show Pythia-14M (a,b) and 70M (c,d); columns show final validation loss and full-model latency versus  $S_{\text{model}}$ . Each row contains 22 checkpoints: the Base Model ( $T_0/P_0$ ), GeLU  $\rightarrow$  ReLU at the hidden FFN activation ( $h$ ) ( $T_1/P_0$ ), and  $T_4/T_7$  thresholding at four/seven sites with pressure applied either at  $h$  ( $P_h$ , dashed) or at all thresholded sites ( $P_{\text{all}}$ , solid). Figure 2 and Table 1 describe the intervention sites. The  $T/P$  curves connect separate runs with parameters controlling the degree of sparsity, while dotted post-hoc paths clip the fixed controls for  $p = 0, .1, \dots, .9$ . Vertical dashed lines mark the analytic sparsity ceilings.

**Execution gains depend on placement.** In a 14M ablation, separate  $h$  and  $z$  skipping saves 150 and 29  $\mu\text{s}$  per sequence, while joint  $qk/pv$  skipping increases latency (Table 2). These results support co-designing training and kernels around exploitable zero structure.

## 2 RELATED WORK

### 2.1 SPARSIFICATION INTERVENTIONS

Post-hoc thresholding removes small activations at inference-time of a pre-trained model. While CATS thresholds small-magnitude activations within FFNs (Lee et al., 2024), TEAL extends it to attention (Liu et al., 2024). These studies motivate training-time interventions to adapt activation distributions to sparse computation and improve the trade-off between quality and sparsity.

ReLU Strikes Back studies the effect of activations functions in language models (Mirzadeh et al., 2023), showing clear sparsity gains on the use of ReLUs. ProSparse combines progressive  $\ell_1$  pressure and thresholding at FFNs during continued training (Song et al., 2024). It does not, however, extend its interventions to attention and restrict the analysis to local FFN sparsity. Cetin et al. (2026) extends ProSparse setup to pre-training and propose specialized kernels. These studies show that training can reshape the quality-sparsity trade-off, but they’re restricted to a single local intervention at FFN sites.

Q-Sparse extend sparsification beyond FFNs. It applies top- $k$  selection to both attention and FFN projection inputs (Wang et al., 2024), but with no  $\ell_1$  pressure. Its top- $k$  operation introduces, however, additional overhead due to the sorting operation. Spark Transformer tries to address it with statistical top- $k$  (You et al., 2025), but we take a simpler approach and use a constant thresholding non-linearity. We argue that when the intervention is introduced during training, the network can adapt itself. Additionally, we evaluate the complementary effect of regularization pressure on both attention and FFNs.

## 2.2 INFERENCE SPEEDUP

Turning activation sparsity into speedup requires a specialized kernel adapted to the model architecture. TEAL accelerates batch-one autoregressive decoding, reporting up to  $1.8\times$  speedup at 50% sparsity (Liu et al., 2024). Cetin et al. (2026) address batched matrix multiplication with their TwELL kernel, and report  $1.3\times$  speedup for batch inference. Spark likewise couples its sparsity mechanism with an implementation that reports decoding gains (You et al., 2025). In each setting, skipped computation and memory access must compensate the additional overhead introduced by the sparse operations. This development has been increasingly accelerated by the use of coding agents, through iterative kernel development with execution and profiling feedback (Ouyang et al., 2025; Luo et al., 2026). We take a related approach and use GPT-6 Astra to develop a specialized kernel based on our training interventions (OpenAI, 2026).

## 3 METHODOLOGY

### 3.1 SPARSIFICATION INTERVENTIONS

A sparsification intervention specifies where activations are thresholded (thresholding sites  $\mathcal{N}$ ) and where activation pressure is applied (pressure sites  $\mathcal{P}$ ). The sets  $\mathcal{N}$  and  $\mathcal{P}$  need not coincide, and the thresholding and pressure functions may take different forms. Together, these choices define a combinatorial space of training interventions. We restrict this space to keep the computational cost manageable while addressing the central research questions.

For thresholding, we avoid the overhead of exact top- $k$  selection (Wang et al., 2024; You et al., 2025) and instead use one-sided and symmetric fixed-threshold operators  $G_{+, \kappa}(x) = x\mathbf{1}\{x \geq \kappa\}$  and  $G_{\pm, \kappa}(x) = x\mathbf{1}\{|x| \geq \kappa\}$ , with  $\kappa \geq 0$ . The constant  $\kappa$  controls the strength of the intervention and is held fixed within each training condition. At  $\kappa = 0$ ,  $G_{+, \kappa}$  reduces to ReLU, whereas  $G_{\pm, \kappa}$  reduces to the identity.

For activation pressure,  $\ell_1$  averages across targeted sites and layers the mean absolute activation at each site after any thresholding. Ordinary L1 pressure minimizes  $\mathcal{L}_{\text{task}} + \lambda\ell_1$ , where  $\lambda$  controls the pressure intensity (Song et al., 2024; Cetin et al., 2026). To limit conflict between the  $\ell_1$  pressure and the task update while reducing the need to tune an additional hyperparameter, we introduce orthogonal  $\ell_1$  pressure (OL1), inspired by gradient-conflict projection methods (Yu et al., 2020; Hsieh et al., 2024). OL1 adds a separately preconditioned pressure correction after the task-only AdamW update, projects away its component opposing the adaptive task direction, and caps its norm relative to that direction. At a fixed optimizer state, once the norm budget is active, further increasing  $\lambda$  leaves the pressure correction unchanged up to numerical stabilizers. To keep the multisite intervention study tractable, we use a common  $\lambda = 1$  and budget  $b = 1$  and directly measure whether the budget becomes active during training. Appendix A.1 gives the full formulation, a brief comparison and shows that budget saturation depends strongly on the target set.

Figure 2 locates the intervention sites within the Pythia architecture. The variables  $a$  and  $m$  denote the normalized inputs to the attention and FFN branches, respectively;  $h$  is the FFN hidden activation;  $z$  is the concatenated attention context before the output projection; and  $q$ ,  $k$ , and  $v$  are the query, key, and value operands. Attention probabilities are denoted by  $p$ . We apply  $G_{+, \kappa}$  at selected sites in  $\{a, m, h, z\}$  and  $G_{\pm, \kappa}$  at selected sites in  $\{q, k, v\}$ . When  $h$  is targeted, the thresholding nonlinearity replaces GELU.

### 3.2 SPARSITY MEASURE AND ARCHITECTURAL REACH

*Activation sparsity* is the fraction of zero activations at a site. We define *model-wide sparsity* as  $\mathcal{S}_{\text{model}} = N_0/N_{\text{model}}$ , where  $N_0$  counts scalar multiplications with a zero activation operand in Pythia-block matrix products, and  $N_{\text{model}}$  counts all scalar multiplications in those blocks and the final output projection. We aggregate counts over validation data, excluding masked attention pairs.

To quantify intervention potential a priori, we define the *model-wide sparsity ceiling* as  $\mathcal{S}_{\text{model}}^{\text{max}} = N_{\text{reach}}(\mathcal{N})/N_{\text{model}}$ , where  $N_{\text{reach}}(\mathcal{N})$  counts scalar multiplications guaranteed a zero activation operand if all selected thresholding sites are 100% sparse. Thus,  $\mathcal{S}_{\text{model}}^{\text{max}}$  is the maximum sparsity di-

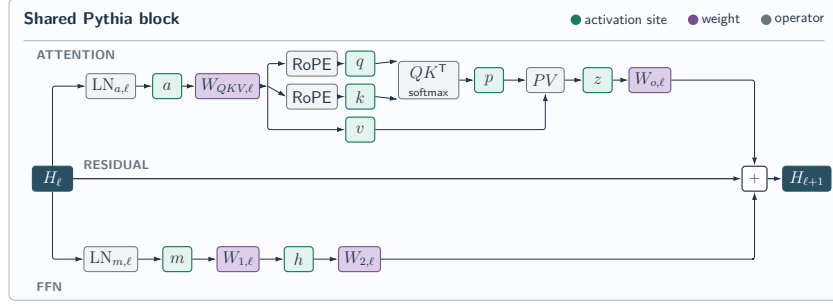


Figure 2: **Intervention sites in a shared Pythia block.** The attention and FFN branches use separate LayerNorms and join the residual path. Green nodes denote activation sites, and purple nodes denote learned weights. Table 1 specifies the thresholding and pressure assignments.

rectly attributable to  $\mathcal{N}$ . It depends on architecture, selected sites, and sequence length. Appendix B gives the precise accounting and definitions.

#### 4 EXPERIMENTAL STUDY

We pretrain Pythia-14M, 70M, and 410M from random initialization (Biderman et al., 2023) on MiniPile (Kaddour, 2023), using 1.493 billion input tokens per model and training for one full-pass. We evaluate validation loss and  $\mathcal{S}_{\text{model}}$  over all validation sequences. Table 1 defines the thresholding and pressure recipes and their sparsity ceilings across model sizes. It shows how the attainable sparsity ceiling increases with model size, under the same intervention recipe and fixed sequence length.

**Experiments across model sizes.** The 14M runs provide the most extensive controlled ablation, used to identify threshold and pressure regimes. The 70M runs test whether these regimes persist in a larger model. We also attempted to extend this analysis to 410M, but under the same token budget and number of optimization steps, this leaves the model substantially less trained. The baseline run with no intervention had a higher final loss and larger late-training gradient norm, which is evidence of a different training regime. We therefore treat the 410M result as a limited, partial scale-up, and focus the detailed intervention analysis on the 14M and 70M runs.

Within each model size and intervention, runs share initialization, data order and training budget. Appendix C.3 gives the full protocol. For comparison, we apply TEAL-style post-hoc magnitude thresholding (Liu et al., 2024) at  $a, m, h, z$ , with percentile targets  $p = 0, 0.1, \dots, 0.9$  and site-layer thresholds calibrated separately on ten training blocks.

Table 1: **Thresholding, pressure and sparsity ceilings.**  $T_1$  replaces GeLU with ReLU at  $h$ . Pressure uses OL1:  $P_1 = P_h$  targets  $h$ , and  $P_{all}$  matches the thresholding sites ( $\mathcal{P} = \mathcal{N}$ ). For  $T_4/T_7$ , threshold strength is swept over  $\kappa \in \{0, 0.01, 0.05, 0.1, 0.5\}$ , with OL1 pressure weight  $\lambda = 1$  and task-relative norm budget  $b = 1$  fixed. Only the 14M  $T_1/P_1$  ablation sweeps  $\lambda \in \{0.05, 0.1, 0.5, 1\}$  at  $b = 1$ . Ceilings are  $100 \cdot S_{model}^{max}$  at 2,048 tokens; executed recipes by model size are listed in Appendix C.3.

| Recipe                 | Threshold sites $\mathcal{N}$ | Pressure sites $\mathcal{P}$ | Ceiling (%) |       |       |
|------------------------|-------------------------------|------------------------------|-------------|-------|-------|
|                        |                               |                              | 14M         | 70M   | 410M  |
| $T_0/P_0$ (Base Model) | $\emptyset$                   | $\emptyset$                  | 0.00        | 0.00  | 0.00  |
| $T_1/P_0$              | $h$                           | $\emptyset$                  | 4.28        | 12.35 | 24.93 |
| $T_1/P_1$              | $h$                           | $h$                          | 4.28        | 12.35 | 24.93 |
| $T_4/P_0$              | $a, m, h, z$                  | $\emptyset$                  | 12.83       | 37.06 | 74.78 |
| $T_4/P_h$              | $a, m, h, z$                  | $h$                          | 12.83       | 37.06 | 74.78 |
| $T_4/P_{all}$          | $a, m, h, z$                  | $\mathcal{N}$                | 12.83       | 37.06 | 74.78 |
| $T_7/P_0$              | $a, m, h, z, q, k, v$         | $\emptyset$                  | 29.95       | 49.42 | 87.25 |
| $T_7/P_h$              | $a, m, h, z, q, k, v$         | $h$                          | 29.95       | 49.42 | 87.25 |
| $T_7/P_{all}$          | $a, m, h, z, q, k, v$         | $\mathcal{N}$                | 29.95       | 49.42 | 87.25 |

Quality-sparsity trade-offs on Pythia 14M and 70M

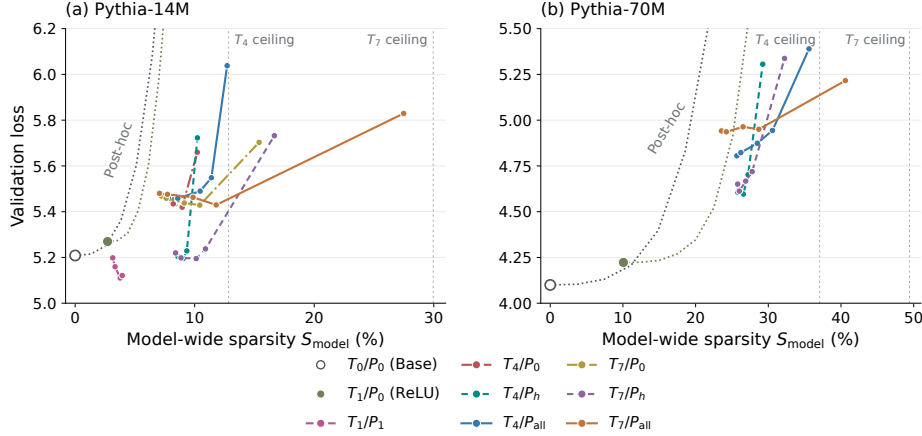


Figure 3: **Quality–sparsity trade-offs at 14M and 70M.** 36/22 trained conditions, with loss scales fitted separately to each size. Colors distinguish executed recipes; lines connect separate threshold settings ( $\kappa$ ), or pressure weights ( $\lambda$ ) for  $T_1/P_1$ . Dotted paths apply post-hoc clipping at  $a, m, h, z$  to the fixed base and ReLU controls. Vertical guides mark the  $T_4$  and  $T_7$  analytic sparsity ceilings.

**Train-time interventions extends the attainable sparsity range, close to the architectural ceiling for the more aggressive recipes, but at a quality cost.** Figure 3 summarizes 58 trained conditions at 14M and 70M. At  $\kappa = 0.5$ ,  $T_7/P_{all}$  reaches 27.48% and 40.60% model-wide sparsity: 91.8% and 82.2% of the respective ceilings. Validation loss increases by 0.62 and 1.12 relative to the same-size base model. At intermediate  $\kappa \in 0.01, 0.05$ , local pressure at  $h$  present a comparable sparsity at lower loss, compared to global pressure  $P_{all}$ . At Section 4.2 we do a comparative analysis on the effect of global vs. local pressure.

#### 4.1 FROM SPARSITY TO SPEEDUPS

**Agentic kernel development.** To test whether learned sparsity reduces inference latency, we used Codex with GPT-6 Astra (OpenAI, 2026), to develop specialized GPU kernels under human guidance. We started from the kernels released by Cetin et al. (2026), although they required adaptation

for Pythia architecture. The agent proposed, implemented and tested the kernels speed and accuracy. During the development phase, we sampled a subset from the training set and tested all kernel changes on them, comparing the model’s output without the custom GPU implementation. We rejected kernels that provided a different model output response. Once we found a kernel that provided speedup for a single sparse model with the same response, we tested it for each training recipe. For each model, we measured inference time and output correctness over the validation set. Its speedup was recorded and included only if these checks passed. We then repeated this evaluation at 70M after adapting the kernels to its larger matrix dimensions. Although we are not specialized in GPU-kernel design, this workflow made direct execution tests practical. This case illustrates how coding agents can support systems development.

**Sparsity and speedup increase together within recipes.** For the implemented kernel, Figure 4 shows an approximately linear increase within each trained threshold/pressure family over the tested grid, although with different slopes across families. At 14M and  $\kappa = 0.5$ , for example,  $T_7/P_{\text{all}}$  is much sparser than  $T_4/P_{\text{all}}$  (27.48% versus 12.71%), yet slightly slower ( $1.38\times$  versus  $1.42\times$  base-model speedup). Although they correlate, a larger  $S_{\text{model}}$  does not directly translate into a larger speedup. The main requirement is the place and structure of zeros and wheater the kernel can exploit them. We investigate it on Section 4.3.

**For exploitable structure, scale increases the opportunity.** Figure 4 shows that the attainable full-model speedup increases with model size. Larger models combine larger matrix operations with higher sparsity ceilings, making sparse computation more effective at reducing end-to-end latency. Accordingly, the best observed speedup rises from  $1.42\times$  at 14M to  $2.08\times$  at 70M.

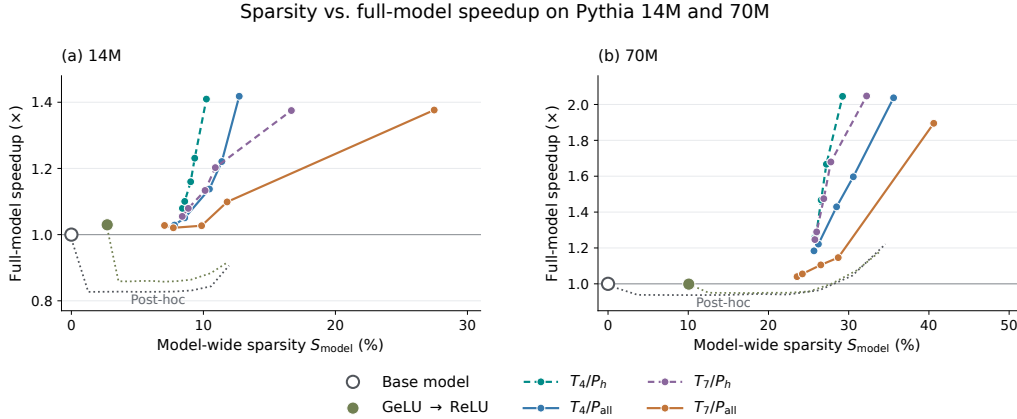


Figure 4: **From model-wide sparsity to measured speedup.** Pythia-14M (a) and 70M (b), with 22 trained checkpoints and 20 post-hoc clipping settings per size. Speedup divides the same-size base-model latency by each setting’s latency, using the final specialized implementation for both terms. Timings use RTX5090, BF16, batch one, 2,048 tokens and the full logit head. Sparsity uses full-validation pooled logical counts.

**Post-hoc clipping does not yield comparable gains in this implementation.** Every post-hoc clipping point remains below  $1\times$  at 14M, and the first degradation point show the additional overhead cost. At 70M, aggressive clipping can exceed  $1\times$ , reaching  $1.22\times$ , but with substantial loss degradation. Here clipping uses a separate masking operations around the kernel, adding additional overhead because it has to iterate over activations on every forward pass. The resulting zeros still have to form groups that the kernel skipping rules can use. The overhead can outweigh the avoided products. However, we did not optimize a fused clipping implementation, so this result does not establish a general limitation of post-hoc sparsification, but **points into a requirement of pairing the sparse intervention with specialized implementation.**

#### 4.2 PAIRED EFFECTS OF GLOBAL VS. LOCAL PRESSURE

One question raised during sparse intervention design is wheater **activation pressure should be applied over the entire network, or are there local sites more suceptible to sparsification?** To

answer this, we compare thresholding across multiple sites ( $T_4$  and  $T_7$ ) with global and local FFN hidden pressure ( $P_{all}$  and  $P_h$ ) at Figure 5.

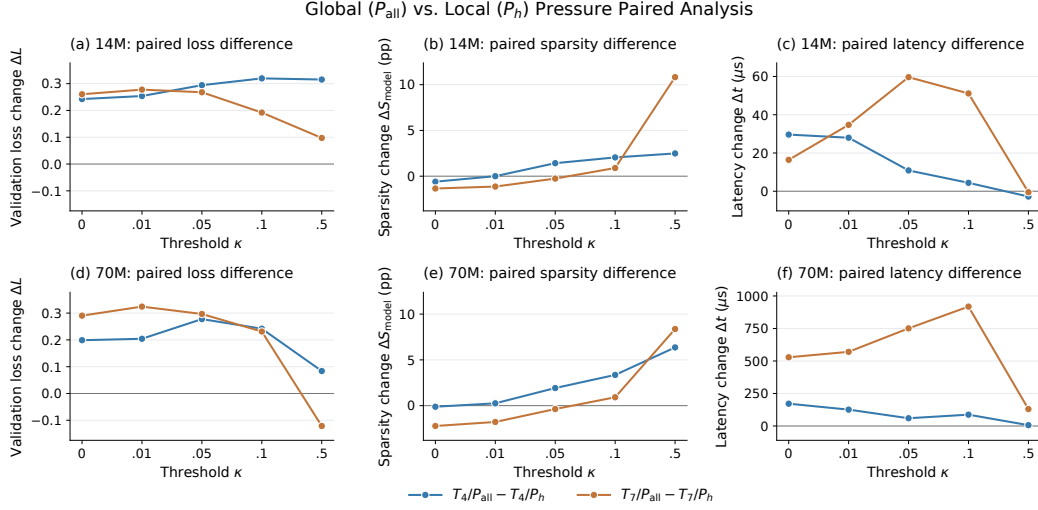


Figure 5: **Paired effects of broadening pressure from  $P_h$  to  $P_{all}$ .** Every point is  $P_{all}$  minus  $P_h$  at matched model size, threshold sites and  $\kappa$ , with  $\lambda = b = 1$ . Rows show 14M/70M; columns show loss change, sparsity change (pp), and full-model latency change ( $\mu s$ ). Positive loss or latency differences favor  $P_h$ ; positive sparsity differences favor  $P_{all}$ .

**Global pressure achieves higher sparsity at a greater quality cost, without translating this additional sparsity into lower latency.** At moderate thresholds ( $\kappa \leq 0.1$ ), local h-only pressure ( $P_h$ ) provides a more favorable quality–sparsity trade-off than global pressure ( $P_{all}$ ) in both model sizes:  $P_h$  lowers loss in all 16 matched comparisons at 14M and 70M (Figure 5), with reductions of 0.19–0.32 while sparsity differs by at most 3.36 percentage points. Although broader pressure can achieve higher sparsity, this gain does not translate into lower measured latency: h-only pressure is faster in 18 of the 20 matched pairs, including all ten 70M pairs. Thus, applying sparsity pressure globally trades model quality for additional sparsity without yielding a corresponding inference-time benefit.

**Local pressure can produce nonlocal sparsity changes.** Figure 6 present the sparsity by site and layer across  $T_4$  and  $T_7$  interventions, for all pressures at 14M. At  $\kappa = 0.05$ ,  $P_h$  increases sparsity at both  $h$  and  $z$  relative to  $P_0$ , although  $z$  receives no direct pressure. Under  $T_7$ ,  $P_h$  also produces higher sparsity at  $h$  and  $z$  than  $P_{all}$ , despite the latter targeting both sites. One possible explanation is that  $(q, k, v)$  increases  $\ell_1$  pressure norms, and  $P_{all}$  caps the OL1 norm more frequently (Appendix A.1), reducing the local at  $h$ . The takeaway is that a local site pressure may spillover and affect the distributions of other sites.

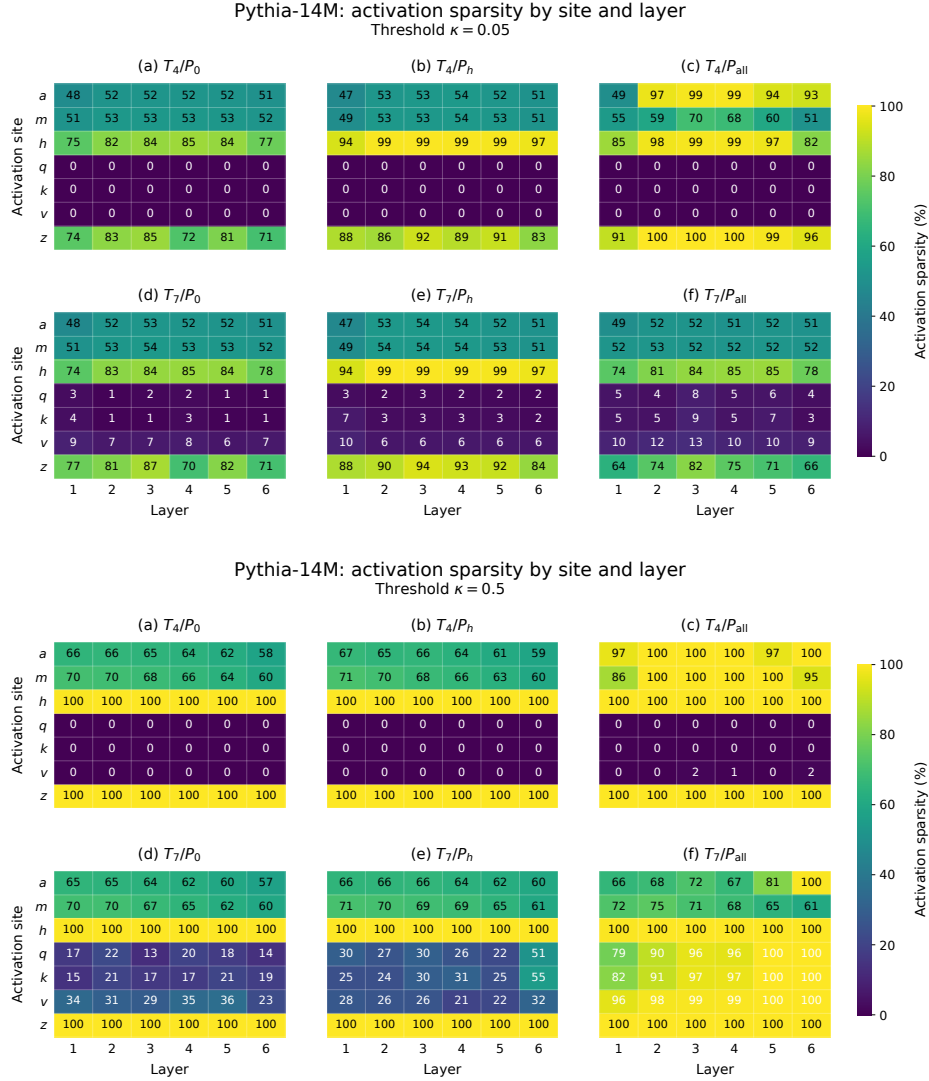


Figure 6: **Activation sparsity by site and layer in Pythia-14M.** The six threshold/pressure recipes at  $\kappa = 0.05$  (top) and  $0.5$  (bottom). Each cell gives the percentage of zero activations over all complete validation sequences at one site and layer. Q/K are post-RoPE; all panels share a 0–100% color scale. Values are rounded to whole percentages, so 0 or 100 does not establish a completely dense or all-zero tensor.

**(h, z) are more susceptible to sparsification.** At higher threshold ( $\kappa = 0.5$ ) (h,z) sites become fully sparse, even if no pressure is applied; while (q,k,v) sites are in the [14, 36] and (a,m) are in [57, 70] sparsity range. This implies the distribution of (q,k,v) and (a,m) may have different ranges and require different threshold values than (h,z).

### 4.3 WHEN SPARSITY TRANSLATES TO SPEEDUP

Sparsity reduces latency when avoided work exceeds the overhead cost of detecting and skipping it. At h, z, the sparse kernel checks  $8 \times 16$  activation tiles before loading their associated weights, so empty tiles avoid both weight load from memory and the matrix multiplications. This strategy, however, doesn't extend to other sites. Applying it at a and m increased full-model latency by 17.2% and 27.4% on 14M  $T_7/P_{all}$  at  $\kappa = 0.5$ , with unchanged outputs. Only 11.12% of a's  $8 \times 16$  tiles

were empty, and none at  $m$ , so most groups still required matrix multiplication and the additional overhead was not justified.

Attention follows FlashAttention’s tiled sequence (Dao et al., 2022): load  $q, k$ , compute scores, apply masking and softmax, then multiply probabilities by  $v$ . Our implementation add additional overhead after the checks, so skipping an empty tile’s multiplication cannot recover the latency from reading from memory. Softmax still runs and turns zero scores into nonzero probabilities, so  $q/k$  sparsity does not automatically propagate to  $pv$ . Zeros in  $v$  remain usable, and sparsifying the probabilities is a further untested opportunity.

We isolate each site’s effect at 14M by disabling it during inference time. Table 2 reports the *bypass* (i.e. percentage of skipped matrix instructions) and saved time per-site sparse operation for the sparser 14M recipe  $T_7/P_{\text{all}}, \kappa = 0.5$ . It makes clear that savings come from  $h$  (150  $\mu\text{s}$ ) and  $z$  (29  $\mu\text{s}$ ). Individual  $a, m, q, k$  effects remain unresolved against process variation, whereas  $pv$  skipping adds latency. Enabling  $qk/pv$  skipping together raises full-model latency from 0.514 to 0.521 ms despite substantial bypass. Thus, for this checkpoint and implementation, attention’s arithmetic savings do not offset its overhead, while  $h/z$  benefit from avoiding weight reads as well as computation.

Table 2: **Conditional latency savings:** 14M  $T_7/P_{\text{all}}, \kappa = 0.5$ . Latency is measured per 2,048-token sequence on RTX 5090 (BF16, batch one). Saved time is full-model latency with one skipping path off minus latency with all paths on; positive values indicate a benefit (reduced latency). Brackets give [minimum, maximum] savings across all off/on pairs of three independent timing runs. Each run reports the geometric mean over 64 sequences.

| Operation                         | Bypass (%) | Saved time ( $\mu\text{s}$ ) | Process span ( $\mu\text{s}$ ) |
|-----------------------------------|------------|------------------------------|--------------------------------|
| $a$ : QKV projection              | 8.4        | −1.6                         | [−6.0, +3.8]                   |
| $m$ : FFN-up                      | 0.0        | −2.4                         | [−5.1, +2.7]                   |
| $h$ : FFN-down                    | 94.5       | +150.1                       | [+145.4, +154.4]               |
| $z$ : attention-output projection | 99.6       | +29.0                        | [+23.7, +35.7]                 |
| $q, k$ : $QK$ scores              | 58.0       | −2.9                         | [−7.3, +1.1]                   |
| $v$ : $PV$ product                | 67.2       | −6.2                         | [−9.0, −2.2]                   |

## 5 DISCUSSION AND CONCLUSION

Our results suggest that sparsification should be understood as a joint training-and-deployment problem rather than as a property of activation distributions alone. First, **local pressure does not imply a local response**: applying pressure only at the FFN hidden site can alter sparsity at other sites, showing that the network can reorganize its internal representations in response to a localized training intervention. More broadly, **sparsity is useful only if the resulting zero structure can be exploited by the inference system**.  $S_{\text{model}}$  is best viewed as a training-time proxy for computational opportunity, not as a direct predictor of latency. In our measurements, the latency gains comes from ( $h, z$ ) sites, which were the ones more susceptible to sparsification, with higher achieved sparsity. One hypothesis is that we must achieve a high ( $> 90\%$ ) level of site-layer sparsity, so the zeros gets a exploitable tiled structure with higher probability.

Transformers appear sufficiently flexible to adapt their internal representations to architectural and optimization constraints, suggesting that larger gains may come from **designing training objectives around inference requirements**, rather than first maximizing a generic sparsity metric and only afterward attempting to accelerate it. The same principle may extend beyond activation sparsity: training could be used to shape representations for properties required by weight sparsity, quantization, batched execution, or other hardware-specific constraints. In this view, deployment requirements are not merely constraints imposed after training, but potential signals for deciding what properties the network should learn.

Several limitations define the scope of these conclusions and point to concrete directions for further work. Our scaling evidence is limited: although the 410M experiments demonstrate that high model-wide sparsity remains attainable, the fixed token budget leaves 410M in a substantially different optimization regime, so it should be interpreted as a stress test rather than a clean scaling experiment.

Our runtime study is also restricted to batch-one, full-sequence inference on an RTX5090; we do not study how sparsity interacts with batched inference, cached decoding, or other hardware.

In addition, the full intervention space is combinatorial, and under the available compute budget we evaluate only a selected set of threshold and pressure placements rather than an exhaustive factorial study. The nonlocal response visible in Figure 6 suggests one particularly promising direction: interventions concentrated at  $h$  and  $z$  may preserve much of the execution benefit while avoiding pressure or thresholding at sites whose sparsity is expensive in quality or unprofitable for the current kernel. Related extensions include learning or tuning different thresholds  $\kappa$  at different sites and directly sparsifying attention probabilities after softmax, which could expose work in  $pv$  without relying on  $q/k$  zeros to propagate through softmax. More generally, rather than asking where the model can be made sparse, a productive systems-guided approach is to decompose full-model inference into individual operations, identify where latency and removable work are concentrated, determine what structured representation would allow those operations to be skipped efficiently, and then design training-time interventions that encourage precisely that structure.

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## A INTERVENTION DEFINITIONS

### A.1 ACTIVATION PRESSURE AND OL1

**Objective and update.** For  $J$  activation tensors  $A_j$  at the sites and layers targeted by  $\mathcal{P}$ , each with  $n_j$  entries after any thresholding, define

$$\ell_1(\theta; \mathcal{P}) = \frac{1}{J} \sum_{j=1}^J \frac{1}{n_j} \sum_{i=1}^{n_j} |A_{j,i}|. \quad (1)$$

Each tensor has equal weight, so changing  $\mathcal{P}$  changes both the targets and normalization. Ordinary L1 applies AdamW (Loshchilov & Hutter, 2019) to the globally clipped gradient of  $\mathcal{L}_{\text{task}} + \lambda \ell_1$ . OL1 separates task and pressure updates: task-only AdamW is followed by a pressure correction that removes the opposing component and limits its norm. Pressure therefore does not directly enter Adam’s moment estimates.

Both gradients are evaluated at the same parameters and averaged over the same microbatches. The clipped task gradient updates AdamW’s bias-corrected moments  $\hat{m}_t, \hat{v}_t$ ; the pressure gradient  $g_1 = \nabla_{\theta} \ell_1$  is unclipped and has no momentum. Using the same preconditioner for both directions gives

$$u = \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon_A}, \quad w = \frac{g_1}{\sqrt{\hat{v}_t} + \epsilon_A}. \quad (2)$$

Norms and inner products include parameters with both gradients and initialized Adam state. For  $c = \langle u, w \rangle$ ,  $q = \|u\|_2^2$  and stabilizer  $\epsilon > 0$ ,

$$\tilde{w} = \begin{cases} w - \frac{c}{q + \epsilon} u, & c < 0 \text{ and } q > \epsilon, \\ w, & \text{otherwise.} \end{cases} \quad (3)$$

Aligned pressure is retained. For weight  $\lambda \geq 0$  and relative norm budget  $b > 0$ ,

$$r = \frac{\lambda \|\tilde{w}\|_2}{\|u\|_2 + \epsilon}, \quad s = \min \left\{ 1, \frac{b}{r + \epsilon} \right\} \quad (r > 0), \quad (4)$$

with  $s = 1$  when  $r = 0$ . After task-only AdamW, each parameter group  $p$  receives  $\theta_p \leftarrow \theta_p - \alpha_p \lambda s \tilde{w}_p$ , where  $\alpha_p$  is its learning rate. Before learning-rate scaling and ignoring stabilizers, the correction norm is  $\min\{\lambda \|\tilde{w}\|_2, b \|u\|_2\}$ . At a fixed optimizer state, increasing  $\lambda$  has no further effect once the cap is active.

OL1 combines conflict projection (Yu et al., 2020) and a smoothed task direction (Hsieh et al., 2024) in Adam’s preconditioned coordinates. The stabilizer makes orthogonality approximate. This calculation excludes group learning rates and decoupled weight decay; avoiding opposition to the Adam direction does not guarantee unchanged task or validation loss.

**L1 versus OL1.** At 14M  $T_1/P_1$ , matched initialization, data order and training budget yield similar results (Table 3). Sparsity differs by less than 0.02 percentage points at every  $\lambda$ . OL1 has slightly lower loss at  $\lambda \in \{0.05, 0.1, 0.5\}$  and higher loss at  $\lambda = 1$ . One seed cannot establish whether these small loss differences persist across seeds.

Table 3: **L1 and OL1 at  $T_1/P_1$  on Pythia-14M.** Final validation loss and model-wide sparsity after 712 matched training steps; OL1 uses  $b = 1$ . Both metrics cover all 338 complete validation blocks (500 documents; 1,444 tail tokens excluded). One initialization per condition.

| $\lambda$ | L1                     |           | OL1                    |           |
|-----------|------------------------|-----------|------------------------|-----------|
|           | $S_{\text{model}}$ (%) | Val. loss | $S_{\text{model}}$ (%) | Val. loss |
| 0.05      | 3.1416                 | 5.2061    | 3.1452                 | 5.1981    |
| 0.1       | 3.3363                 | 5.1655    | 3.3386                 | 5.1594    |
| 0.5       | 3.7876                 | 5.1127    | 3.7680                 | 5.1102    |
| 1         | 3.9493                 | 5.1023    | 3.9384                 | 5.1210    |

**Update geometry and budget saturation.** We measure  $\rho_{\text{opp}} = \max(0, -c)/(q + \epsilon)$ , the removed component’s norm relative to  $\|u\|_2$  before capping. All recorded steps satisfy  $q > \epsilon$ ; aligned steps contribute zero. Figure 7 shows  $\rho_{\text{opp}}$  and  $r/b$  for the 20 14M OL1 conditions in Figure 1, using all 712 updates per condition at  $\lambda = b = 1$ . The cap is active ( $s < 1$ ) on 96.99% of  $T_7/P_{\text{all}}$  steps, compared with 0.51% for  $T_4/P_{\text{all}}$  and 0.25% for either  $P_h$  recipe. The diagnostics are consistent with stronger pressure–task conflict when attention sites are included, but do not isolate each site’s contribution.

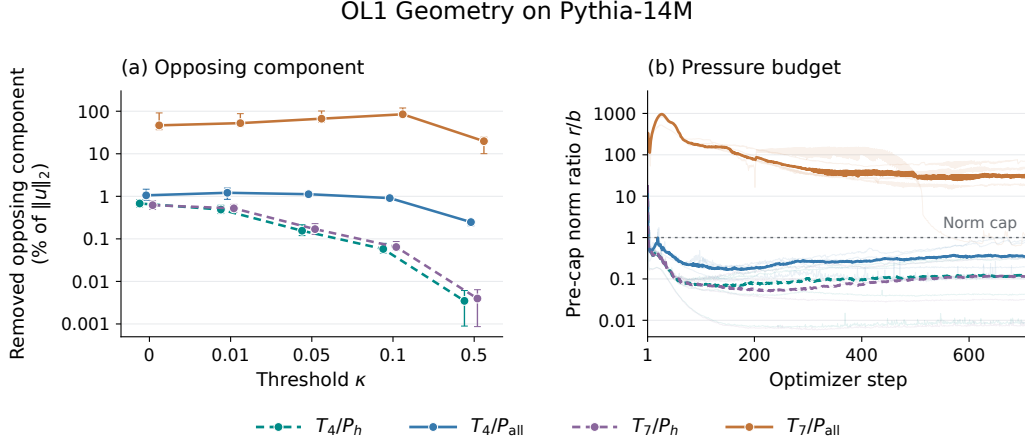


Figure 7: **OL1 geometry depends on pressure placement at 14M.** (a) Median removed opposing component,  $100\rho_{\text{opp}}$ ; bars span the 25th–75th percentiles over 712 training steps per condition, not confidence intervals. (b) Pre-cap norm ratio  $r/b$ : faint curves show individual thresholds, and bold curves their pointwise median. The dotted line marks the norm cap. Colors and line styles match Figure 1; both y-axes are logarithmic, and  $\lambda = b = 1$  throughout.

## B SPARSITY ACCOUNTING

### B.1 ACTIVATION AND MODEL-WIDE SPARSITY

Activation sparsity is the fraction of zero entries at a site, pooling counts across layers and validation sequences before dividing.

Model-wide sparsity counts scalar multiplications with a zero activation input:

$$\mathcal{S}_{\text{model}} = \frac{N_0}{N_{\text{model}}}. \quad (5)$$

$N_0$  counts these multiplications within Pythia blocks;  $N_{\text{model}}$  counts all multiplications in those blocks and the final vocabulary projection. The latter contributes only to the denominator. Counts are pooled across layers and validation sequences; percentages are  $100\mathcal{S}_{\text{model}}$ .

For a block’s linear projection from  $d_{\text{in}}$  to  $d_{\text{out}}$  features over  $n$  rows, the denominator contribution is  $nd_{\text{in}}d_{\text{out}}$ . Each zero input contributes  $d_{\text{out}}$  to  $N_0$ , one multiplication per output feature. Zero weights do not count as activation sparsity.

In attention, both operands are activations. We count each multiplication once if either or both are zero. For queries  $q$ , keys  $k$ , values  $v$  and attention probabilities  $p$ , one head in one sequence contributes

$$N_{0,\text{QK}} = \sum_{t=1}^T \sum_{u=1}^t \sum_{j=1}^{d_h} \mathbf{1}\{q_{tj} = 0 \vee k_{uj} = 0\}, \quad (6)$$

$$N_{0,\text{PV}} = \sum_{t=1}^T \sum_{u=1}^t \sum_{j=1}^{d_h} \mathbf{1}\{p_{tu} = 0 \vee v_{uj} = 0\}. \quad (7)$$

Here  $T$  is sequence length,  $d_h$  the head dimension,  $t$  the query position and  $u$  the key/value position. Restricting  $u \leq t$  excludes masked future positions from both counts. We sum over heads, layers and validation sequences, using the actual operands after thresholding and intervening transformations.

Normalization, softmax, activation functions, additions and memory transfers are excluded, so  $\mathcal{S}_{\text{model}}$  does not directly measure latency or speedup.

## B.2 MODEL-WIDE SPARSITY CEILING

Assume all activations at the thresholding sites  $\mathcal{N}$  are zero. If  $N_{\text{reach}}(\mathcal{N})$  counts the multiplications that must then have a zero activation input, the ceiling is

$$\mathcal{S}_{\text{model}}^{\text{max}} = \frac{N_{\text{reach}}(\mathcal{N})}{N_{\text{model}}}. \quad (8)$$

The denominator is unchanged, and each multiplication is counted once. The numerator includes downstream activations forced to zero: setting  $v = 0$  also makes  $z = 0$ , affecting both  $PV$  and the attention-output projection. Zero attention scores do not imply zero softmax probabilities.

The ceiling depends only on architecture, selected sites and sequence length. Natural zeros outside those sites can raise measured  $\mathcal{S}_{\text{model}}$  above it; the ceiling does not predict the associated quality cost. Appendix C.2 gives the counts for each Pythia size.

## C EXPERIMENTAL SETUP DETAILS

### C.1 PYTHIA ARCHITECTURE AND INTERVENTION SITES

Table 4 gives the shapes and positions of the sites in Figure 2. Here  $T$  is sequence length,  $d$  hidden width,  $d_f$  FFN width and  $H$  the number of heads, each with dimension  $d_h = d/H$ . Sites and threshold are shared across layers.

Table 4: Activation shapes and intervention positions for one sequence.

| Site   | Shape                   | Following matrix product    | Position                       |
|--------|-------------------------|-----------------------------|--------------------------------|
| $a$    | $T \times d$            | QKV projection              | After attention LayerNorm      |
| $m$    | $T \times d$            | FFN up projection           | After FFN LayerNorm            |
| $h$    | $T \times d_f$          | FFN down projection         | Activation-function output     |
| $q, k$ | $H \times T \times d_h$ | Attention scores            | After RoPE                     |
| $v$    | $H \times T \times d_h$ | Probability-value           | After the QKV split            |
| $z$    | $T \times d$            | Attention-output projection | Concatenated attention context |

### C.2 ARCHITECTURE-SPECIFIC SPARSITY CEILINGS

For one sequence, each Pythia block uses  $3Td^2$  scalar multiplications in the QKV projection,  $Td^2$  in the attention-output projection,  $Tdd_f$  in each FFN projection, and  $dT(T+1)/2$  in each of  $QK$  and  $PV$ . With  $L$  layers and vocabulary size  $V_{\text{vocab}}$ , the totals are

$$\begin{aligned} C_{\text{block}} &= T(4d^2 + 2dd_f) + dT(T+1), \\ C_{\text{model}} &= LC_{\text{block}} + TdV_{\text{vocab}}. \end{aligned} \quad (9)$$

Sequence count cancels from the ceiling ratio. For  $d_f = 4d$ , per-layer counts in the ceiling numerator are 0 for  $T_0/P_0$ ,  $4Td^2$  for  $T_1/P_0$  and  $T_1/P_1$ ,  $12Td^2$  for  $T_4$ , and  $C_{\text{block}}$  for  $T_7$ . Multiplying by  $L/C_{\text{model}}$  gives  $\mathcal{S}_{\text{model}}^{\text{max}}$ , independent of pressure placement.

Table 1 uses  $V_{\text{vocab}} = 50,304$  and  $(L, d) = (6, 128), (6, 512), (24, 1,024)$  for 14M, 70M and 410M, respectively, with the common sequence length in Table 5.<sup>1</sup>

<sup>1</sup>Pinned architecture configurations: 14M, 70M, and 410M.

### C.3 TRAINING AND EVALUATION

All sizes use the Pythia-14M-deduped tokenizer and seeds of 1234 for initialization and data order, with one seed per condition. AdamW settings are shared:  $\beta = (0.9, 0.95)$ ,  $\epsilon = 10^{-8}$ , weight decay 0.1 and gradient clipping at norm 1. Attention and hidden dropout are zero. Training uses 712 updates with 1% linear warmup followed by cosine learning-rate decay. Table 5 gives the learning rates, microbatch sizes and shared effective batch. Gradients accumulate over the listed microbatches before each optimizer update.

Table 5: **Training settings across model sizes.** Effective batch size equals microbatch size times gradient accumulation. The common training budget gives fewer tokens per parameter at larger sizes.

| Setting                              | 14M        | 70M        | 410M               |
|--------------------------------------|------------|------------|--------------------|
| Parameters                           | 14,067,712 | 70,426,624 | 405,334,016        |
| Peak learning rate                   | $10^{-3}$  | $10^{-3}$  | $3 \times 10^{-4}$ |
| Minimum learning rate                | $10^{-4}$  | $10^{-4}$  | $3 \times 10^{-5}$ |
| Microbatch (sequences)               | 32         | 4          | 4                  |
| Gradient accumulation (microbatches) | 32         | 256        | 256                |
| Effective batch (sequences/update)   | 1,024      | 1,024      | 1,024              |
| Sequence length                      | 2,048      | 2,048      | 2,048              |
| Tokens per parameter                 | 106.14     | 21.20      | 3.68               |

All sizes include the  $T_0/P_0$  and  $T_1/P_0$  controls. The 14M study adds  $T_1/P_1$  with L1 or OL1, and  $T_4$  and  $T_7$  with  $P_0$ ,  $P_h$  or  $P_{\text{all}}$ . At 70M,  $T_4$  and  $T_7$  use  $P_h$  or  $P_{\text{all}}$ ; at 410M, only  $P_{\text{all}}$ . Parameter grids are given in Table 1. We evaluate the final checkpoint, without validation-based selection. The main figures measure loss and model-wide sparsity on the same validation sequences (Appendix B).

Figure 8 shows the training trajectories. They do not establish convergence, and the lower learning rates at 410M limit comparisons that attribute loss differences to model size alone.

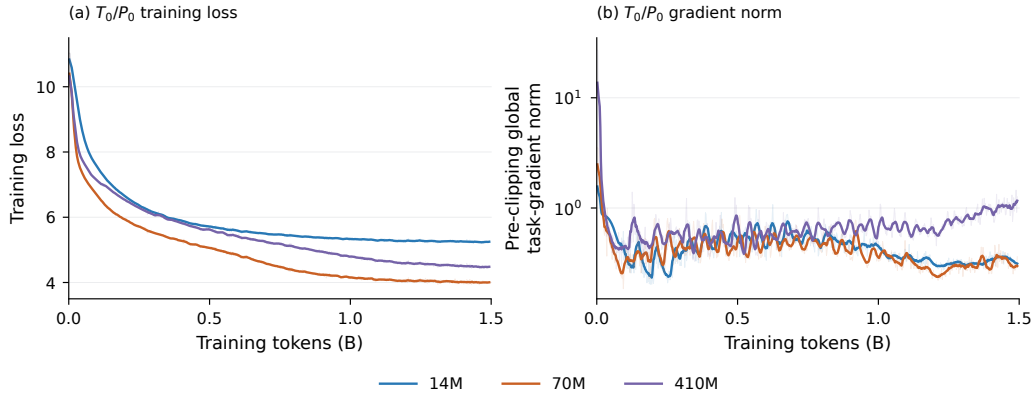


Figure 8: **Base-model ( $T_0/P_0$ ) training across model sizes.** (a) Training loss and (b) gradient norm before clipping (logarithmic scale). Faint lines show individual updates; solid lines show centered nine-update means, with shorter windows at the ends. Gradient norms are computed after accumulation and are not normalized by parameter count.

## D COMPLETE RESULTS AND SUPPORTING DIAGNOSTICS

### D.1 DETAILED TRAINED RESULTS

Tables 6–8 report all 74 trained conditions, including settings outside the favorable quality–sparsity trade-off. Loss and model-wide sparsity use the same definitions and final checkpoints as the main figures, with one training seed and complete validation per condition. Values are rounded.

Table 6: **Additional 14M ablations.** Pressure at  $h$  (top) and thresholding without pressure (bottom), paired by  $\lambda$  or  $\kappa$ . Controls appear in Table 7.

| $\lambda$ | $T_1/P_1$ (L1) |                                  | $T_1/P_1$ (OL1) |                                  |
|-----------|----------------|----------------------------------|-----------------|----------------------------------|
|           | Loss           | $\mathcal{S}_{\text{model}}$ (%) | Loss            | $\mathcal{S}_{\text{model}}$ (%) |
| 0.05      | 5.2061         | 3.142                            | 5.1981          | 3.145                            |
| 0.1       | 5.1655         | 3.336                            | 5.1594          | 3.339                            |
| 0.5       | 5.1127         | 3.788                            | 5.1102          | 3.768                            |
| 1         | 5.1023         | 3.949                            | 5.1210          | 3.938                            |

| $\kappa$ | $T_4/P_0$ |                                  | $T_7/P_0$ |                                  |
|----------|-----------|----------------------------------|-----------|----------------------------------|
|          | Loss      | $\mathcal{S}_{\text{model}}$ (%) | Loss      | $\mathcal{S}_{\text{model}}$ (%) |
| 0        | 5.4705    | 7.212                            | 5.4684    | 7.218                            |
| 0.01     | 5.4665    | 7.414                            | 5.4588    | 7.618                            |
| 0.05     | 5.4341    | 8.206                            | 5.4379    | 9.127                            |
| 0.1      | 5.4196    | 8.953                            | 5.4287    | 10.425                           |
| 0.5      | 5.6597    | 10.216                           | 5.7029    | 15.387                           |

Table 7: **Matched 14M/70M results.** The same threshold/pressure setting at both sizes. Pressure uses OL1 with  $\lambda = b = 1$ .

| Recipe               | $\kappa$ | 14M    |                                  | 70M    |                                  |
|----------------------|----------|--------|----------------------------------|--------|----------------------------------|
|                      |          | Loss   | $\mathcal{S}_{\text{model}}$ (%) | Loss   | $\mathcal{S}_{\text{model}}$ (%) |
| $T_0/P_0$ (Base)     | —        | 5.2086 | 0.000                            | 4.0998 | 0.000                            |
| $T_1/P_0$ (ReLU)     | —        | 5.2696 | 2.714                            | 4.2227 | 10.066                           |
| $T_4/P_h$            | 0        | 5.2157 | 8.409                            | 4.6065 | 25.792                           |
|                      | 0.01     | 5.2045 | 8.586                            | 4.6185 | 25.971                           |
|                      | 0.05     | 5.1956 | 9.036                            | 4.5964 | 26.566                           |
|                      | 0.1      | 5.2287 | 9.343                            | 4.7020 | 27.218                           |
|                      | 0.5      | 5.7227 | 10.227                           | 5.3057 | 29.236                           |
| $T_4/P_{\text{all}}$ | 0        | 5.4582 | 7.810                            | 4.8053 | 25.673                           |
|                      | 0.01     | 5.4583 | 8.587                            | 4.8227 | 26.224                           |
|                      | 0.05     | 5.4898 | 10.456                           | 4.8739 | 28.490                           |
|                      | 0.1      | 5.5483 | 11.394                           | 4.9437 | 30.574                           |
|                      | 0.5      | 6.0380 | 12.713                           | 5.3895 | 35.596                           |
| $T_7/P_h$            | 0        | 5.2198 | 8.399                            | 4.6508 | 25.786                           |
|                      | 0.01     | 5.1981 | 8.856                            | 4.6122 | 26.009                           |
|                      | 0.05     | 5.1950 | 10.126                           | 4.6670 | 26.895                           |
|                      | 0.1      | 5.2374 | 10.904                           | 4.7189 | 27.779                           |
|                      | 0.5      | 5.7320 | 16.664                           | 5.3372 | 32.233                           |
| $T_7/P_{\text{all}}$ | 0        | 5.4802 | 7.054                            | 4.9412 | 23.562                           |
|                      | 0.01     | 5.4758 | 7.723                            | 4.9364 | 24.226                           |
|                      | 0.05     | 5.4628 | 9.863                            | 4.9636 | 26.524                           |
|                      | 0.1      | 5.4295 | 11.797                           | 4.9501 | 28.694                           |
|                      | 0.5      | 5.8294 | 27.483                           | 5.2160 | 40.602                           |

## D.2 POST-HOC CLIPPING AT 14M

Figure 9 extends the base/ReLU comparison in Figure 3 to the trained checkpoints with retained post-hoc sweeps. The detailed view resolves the trade-offs near the trained models; the full-range view shows the loss cost of aggressive clipping. These are changes to fixed models at evaluation, rather than additional training conditions. The clipping protocol is the same as in Section 4. No corresponding sweep was collected for  $T_4/P_h$  or  $T_7/P_h$ .

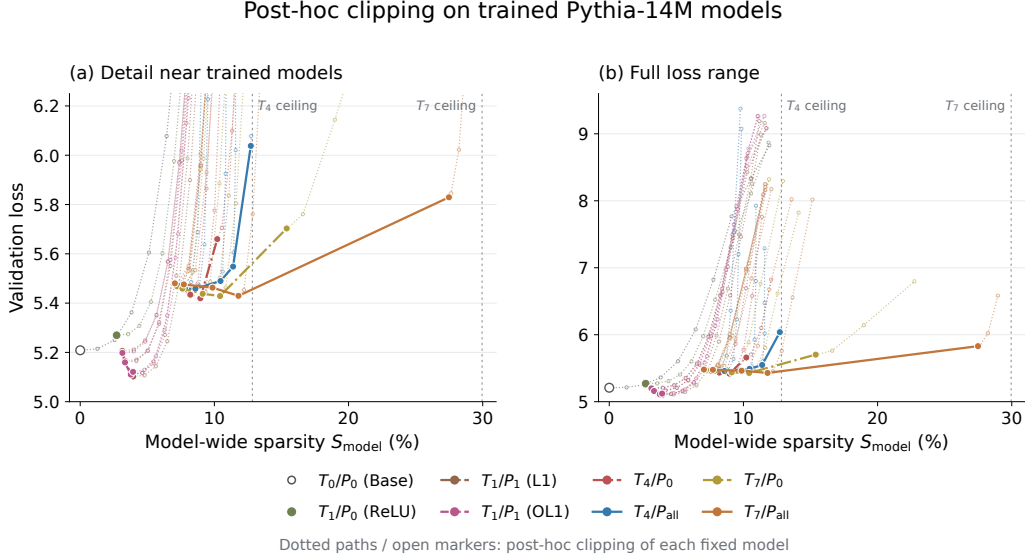


Figure 9: **Post-hoc clipping on trained Pythia-14M models.** Thirty trained checkpoints and ten clipping targets per checkpoint ( $p = 0, 0.1, \dots, 0.9$ ). Both panels show the same settings, with different loss ranges. Prominent markers and recipe lines identify trained models; dotted paths with small open markers follow each fixed model through increasing clipping targets. Colors and recipe labels match the main figures, with the L1 comparison included explicitly. Each path starts at its measured  $p = 0$  point. Lines connect evaluated settings, not fitted frontiers.

## D.3 410M RESULTS

**410M as a fixed-token stress test.** Figure 10 retains all twelve trained 410M conditions and both complete control-clipping paths. At  $\kappa = 0.5$ ,  $T_7/P_{\text{all}}$  reaches 80.62% model-wide sparsity, or 92.4% of the seven-site ceiling, at loss 5.121 versus the base model’s 4.547. This tests attainable sparsity under a larger architecture and the same token budget; it is not a quality-scaling experiment. The 410M seven-site ceiling is 87.25%, compared with 49.42% at 70M, so greater architectural reach also contributes to the larger observed model-wide sparsity.

The 410M base-model loss exceeds 70M’s 4.100. All sizes receive 712 optimizer steps and 1.493B input tokens, corresponding to 106.14, 21.20 and 3.68 tokens per parameter at 14M, 70M and 410M. The 410M schedule also has a lower peak learning rate ( $3 \times 10^{-4}$  rather than  $10^{-3}$ ). These differences may contribute to the non-monotonic quality ordering, but their effects are not isolated. The training trajectories are retained in Figure 8. We therefore treat 410M as a limited stress test and use 14M/70M for detailed intervention and runtime analysis.

Table 8: **410M fixed-token stress test.** All twelve trained conditions; pressure uses OL1 with  $\lambda = b = 1$ . This size has no  $P_h$  or unpressured  $T_4/T_7$  runs.

| Recipe               | $\kappa$ | Loss   | $\mathcal{S}_{\text{model}}$ (%) |
|----------------------|----------|--------|----------------------------------|
| $T_0/P_0$ (Base)     | —        | 4.5475 | 0.011                            |
| $T_1/P_0$ (ReLU)     | —        | 4.6513 | 21.050                           |
| $T_4/P_{\text{all}}$ | 0        | 5.6921 | 38.339                           |
|                      | 0.01     | 5.6914 | 39.149                           |
|                      | 0.05     | 5.6789 | 42.125                           |
|                      | 0.1      | 5.6789 | 45.688                           |
|                      | 0.5      | 5.1911 | 71.591                           |
| $T_7/P_{\text{all}}$ | 0        | 5.4263 | 41.121                           |
|                      | 0.01     | 5.4331 | 41.478                           |
|                      | 0.05     | 5.4679 | 44.053                           |
|                      | 0.1      | 5.4908 | 48.998                           |
|                      | 0.5      | 5.1209 | 80.616                           |

Pythia-410M: fixed-token sparsity stress test

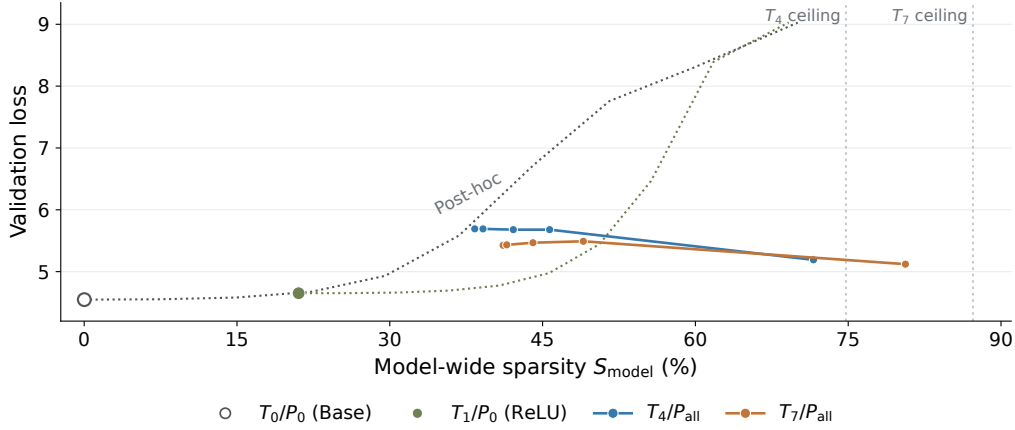


Figure 10: **410M quality-sparsity stress test.** All twelve trained conditions: Base, ReLU, and  $T_4/P_{\text{all}}$  and  $T_7/P_{\text{all}}$  at  $\kappa = 0, .01, .05, .1, .5$ . Dotted paths show all ten post-hoc targets for each fixed Base/ReLU control, including their full loss range. Vertical guides mark the  $T_4$  and  $T_7$  sparsity ceilings.

## E KERNEL DEVELOPMENT AND EVALUATION

**Implementation.** The specialized forward pass preserves each trained model’s weights, thresholds, biases and residual connections. It combines CUDA, CUTLASS and FlashAttention components to fuse operations and skip matrix instructions whose activation operands are zero. It introduces no additional pruning. One kernel computes the two input LayerNorms from their shared input while retaining separate affine parameters and thresholding. QKV layout conversion, RoPE and the selected  $q, k, v$  gates are fused;  $q, k$  gates follow RoPE. The FFN-down and attention-output projections are evaluated jointly with their biases and residual addition. The final vocabulary projection remains dense. Table 9 specifies the sparse execution rules.

The scalar path requires finite weights and biases, with nonzero magnitudes between  $2^{-50}$  and  $2^{50}$ ; selected activations obey the same magnitude bounds. Other rows use the matrix path. The two sizes

Table 9: **Sparse execution rules.** Tile dimensions are tokens by input features. A matrix multiply-accumulate (MMA) instruction operates on  $16 \times 8 \times 16$  elements (output rows, output columns, reduction dimension). Zero checks and any scalar replacement execute during every forward pass.

| Operation                             | Execution rule                                                                                                                                                                | Work avoided                                                                                    |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| QKV, FFN-up ( $a, m$ )                | Test each $16 \times 16$ activation fragment after both operands have been loaded.                                                                                            | Skip the MMA for an empty fragment; retain operand reads.                                       |
| FFN-down, attention output ( $h, z$ ) | Inspect eight rows; use scalar arithmetic for rows with at most two nonzeros when numerical guards pass. Process the remaining rows in $8 \times 16$ tiles padded to 16 rows. | Skip empty tiles before loading their weights. Scalar rows read only the corresponding weights. |
| Attention ( $QK, PV$ )                | After loading the operands, use a warp-wide check to test whether either complete MMA fragment is zero.                                                                       | Skip that MMA; retain input loading, masking, softmax and output processing.                    |

retain these rules, with projection dimensions adapted from  $(d, d_f) = (128, 512)$  to  $(512, 2048)$ . The 14M attention kernel uses  $64 \times 256$  query/key tiles and two key/value partitions; 70M uses  $128 \times 128$  tiles without partitioning to match its native attention reduction order. This change was needed for numerical agreement, rather than selected by a 70M performance search.

**Scope and numerical checks.** Tests use one RTX 5090, BF16 inference, batch size one, 2,048-token causal sequences, and all 50,304 output logits, with PyTorch 2.11.0, Transformers 5.12.1 and CUDA 12.8. The implementation assumes fixed weights, shapes and non-overlapping use of its buffers. It is not evaluated for training, cached decoding, concurrent serving, arbitrary masks or thresholds outside the tested grid. Each final checkpoint must pass in three fresh processes on all 338 validation sequences. Relative to the unmodified eager implementation, outputs must be finite, every logit must satisfy  $|y - \hat{y}| \leq 0.25 + 0.02|\hat{y}|$ , the relative  $\ell_2$  error of each sequence’s logits must not exceed 0.02, and the absolute mean validation-loss difference must not exceed 0.001. These are numerical tolerances, not bitwise equivalence.

**Timing and reference.** Native PyTorch/SDPA and specialized inference use CUDA graphs with identical inputs. Latency is the geometric mean of host timings over 64 fixed validation sequences, seven randomized paired passes and three fresh processes. Compilation, static weight layout preparation, graph capture and equal input staging are excluded. Embeddings, all blocks, final normalization, vocabulary logits and input-dependent preprocessing are included. Post-hoc masks also remain inside timing; no activations are cached. Measurements use separate GPU sessions, so common normalization does not remove session effects.

For comparisons across recipes, Figure 11 uses *one native base-model latency per size*:  $t_{\text{native}, T_0/P_0} / t_{\text{specialized}, r}$  for recipe  $r$ . The native references are 0.65544 ms at 14M and 1.66388 ms at 70M. This differs from Figure 4, which uses the specialized base-model latencies, 0.65157 and 3.31325 ms. The best trained-recipe gains are therefore  $1.427\times$  and  $1.028\times$  over the native base, versus  $1.418\times$  and  $2.047\times$  over the specialized base. The latter 70M ratio reflects a slow dense path in the ported implementation and does not establish a twofold gain over native inference.

**What MMA bypass measures.** Bypass is the number of omitted MMA instructions divided by issued plus omitted instructions, pooling integer counts across layers and validation sequences. Projection totals pool the four projection operations; attention totals pool  $QK$  and  $PV$ . We keep the two groups separate because equal instruction counts need not imply equal costs. At  $h, z$ , bypass includes replacement by scalar arithmetic and counts padded rows; it is not a fraction of all arithmetic eliminated. Attention counts include causal padding; even the base model bypasses 10.42% of  $PV$  instructions at 14M and 5.15% at 70M. In contrast,  $\mathcal{S}_{\text{model}}$  excludes masked pairs and includes the dense vocabulary projection in its denominator.

The figure’s projection scalar sparsity pools zero-activation products over all projection products in the actual BF16 operands. Both this measure and bypass cover all 338 validation sequences; timing uses the 64-sequence subset. Thus these diagnostics complement, rather than redefine, the canonical model-wide sparsity used in the quality figures.

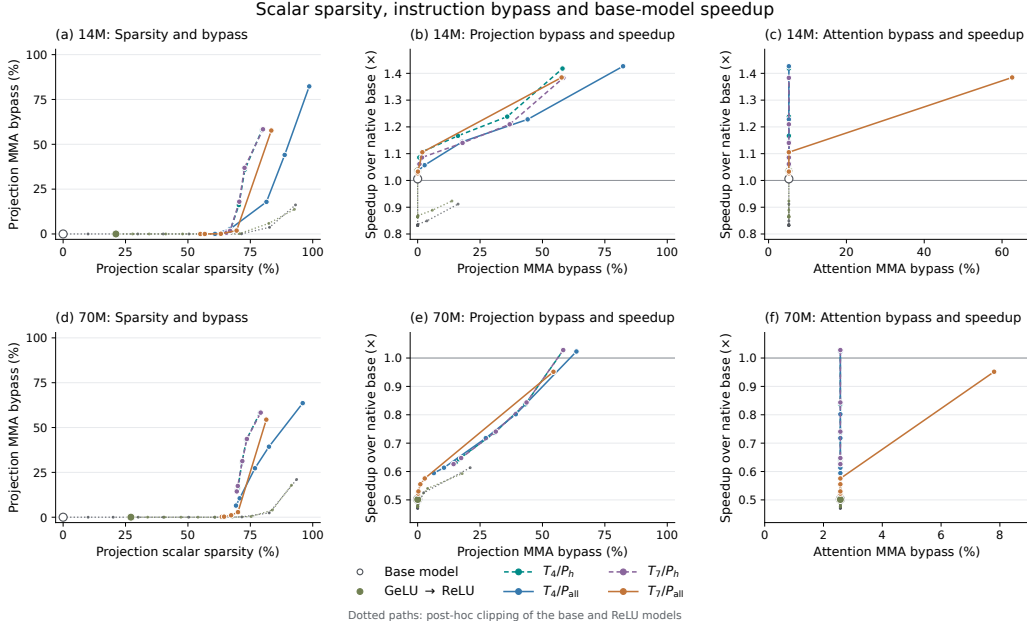


Figure 11: **Scalar sparsity, instruction bypass and speedup over a fixed native base model.** Rows show 14M and 70M, each with 22 trained settings and 20 post-hoc control settings. Left: projection scalar sparsity versus projection MMA bypass. Middle/right: projection/attention bypass versus full-model speedup. Every speedup within a row uses the same native  $T_0/P_0$  reference; the horizontal line is  $1\times$ . The Base model marker uses specialized inference, so it need not lie on that line. Lines connect measured thresholds, not fitted trends; dotted paths are post-hoc clipping. Counts are pooled, include the instruction-padding and scalar-substitution conventions above, and do not measure saved memory traffic.

**Interpretation and controlled tests.** Projection bypass tracks speedup in both sizes, but high scalar sparsity alone need not produce empty instruction tiles. Post-hoc controls show this gap and incur separate masking costs. Attention bypass is nearly constant across the trained grid except for  $T_7/P_{all}$  at  $\kappa = 0.5$ ; this single exception cannot establish a general attention-speedup relation. At 70M it has more attention bypass but less native-base speedup than  $T_7/P_h$  ( $0.952\times$  versus  $1.028\times$ ). Bypass therefore describes exploitable structure without explaining all latency differences.

Table 2 provides the direct 14M test: disable one skipping path while keeping weights, gates and fusion fixed. Its min-max spans cover all pairs of three independent off/on process means; effects are conditional and non-additive. These tests do not separately time memory reads, zero checks or softmax. No corresponding operation-level ablation was run at 70M, so its scatter is descriptive evidence, not causal attribution.

**Alternatives tested.** Porting the  $h, z$  strategy to  $a, m$  was correct but slower in the tested 14M case (Section 4.3). The padded fallback issued  $1.90\times$  and  $2.00\times$  as many matrix instructions, respectively. Attention shortcuts using cumulative value sums for zero queries, disjoint query/key supports, or cheaper conditional execution either failed numerical checks or did not establish a reliable gain. Disabling attention skipping improved all 35 historical 14M comparisons. The retained implementation is therefore not latency-optimal; it preserves one common execution policy across recipes. Earlier detection that avoids attention operand reads, fused post-hoc masking and sparsification of softmax probabilities remain untested here.